affirmoTM tutorial 5: results

The results object, the comparison, the CE plane and the output files

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Table of contents

1 The results object 1

2 Each arm 2

3 The comparison 3

4 The cost-effectiveness plane 4

5 The output files 4

This tutorial starts from a fresh analysis folder with the example input files, and runs the model (see tutorial 1):

```
library(affirmoTM)
library(data.table)  # for fread()
```

```
tm_example_inputs()
results <- tm_run()
```

1 The results object

`tm_run()` returns the results and prints a summary:

```
results
```

```
<tm_results> mean per arm over 100 runs (discounted costs and QALYs)

          arm patients cost_total qaly_total intervention_cost_total
       <char>    <num>      <num>      <num>                   <num>
1:       iabc     5000   65493591   18004.50                187494.6
2: usual_care     5000   66035435   17980.58                     0.0
   cost_per_patient qaly_per_patient intervention_cost_per_patient ischaemic_stroke
              <num>            <num>                         <num>            <num>
1:         13098.72            3.601                        37.499           429.23
2:         13207.09            3.596                         0.000           461.93
   haemorrhagic_stroke major_bleed  deaths
                 <num>       <num>   <num>
1:              121.54      147.47 1850.85
2:              129.50      160.45 1853.00

iABC minus usual care (mean over runs, per patient):
```

```
  Incremental cost:   €-108.37 (intervention cost: €37.5)
  Incremental QALYs:  0.00478
  ICER:               iABC dominant (cheaper, more QALYs)
  Difference in events (whole cohort): ischaemic_stroke -32.7, haemorrhagic_stroke -8.0, major_bleed
   ↪  -13.0, deaths -2.1

Written to outputs/: summary.csv, summary_by_run.csv, comparison.csv, comparison_by_run.csv, ce_plane.png
```

The results are a list:

```
names(results)
```

```
[1] "by_run"            "summary"           "comparison_by_run" "comparison"
[5] "n_runs"            "currency"          "wtp"
```

Element	What it holds
`by_run`	One row per run and arm: the summary of each run
`summary`	The mean over the runs, one row per arm
`comparison_by_run`	iABC minus usual care, one row per run
`comparison`	The mean differences, the ICER and its quadrant
`n_runs`, `currency`, `wtp`	The settings used

2 Each arm

```
results$summary
```

```
         arm patients cost_total qaly_total intervention_cost_total
      <char>    <num>      <num>      <num>                   <num>
1:       iabc     5000   65493591   18004.50                187494.6
2: usual_care     5000   66035435   17980.58                     0.0
   cost_per_patient qaly_per_patient intervention_cost_per_patient ischaemic_stroke
              <num>            <num>                         <num>            <num>
1:         13098.72         3.600901                      37.49893           429.23
2:         13207.09         3.596116                       0.00000           461.93
   haemorrhagic_stroke major_bleed  deaths
                 <num>       <num>   <num>
1:              121.54      147.47 1850.85
2:              129.50      160.45 1853.00
```

Column	Meaning
`patients`	Patients who enter within the horizon
`cost_total`, `qaly_total`	Discounted totals for the whole cohort
`intervention_cost_total`	The discounted intervention cost within `cost_total`
`cost_per_patient`	`cost_total` divided by the number of patients
`qaly_per_patient`	`qaly_total` divided by the number of patients
`intervention_cost_per_patient`	`intervention_cost_total` divided by the number of patients
one column per event	The years spent in that event's states (see `events` in `settings.yaml`)
`deaths`	The number of patients who died

Costs and QALYs are per patient; events and deaths are counts for the whole cohort.

3 The comparison

iABC minus usual care, as the mean over the runs:

```
results$comparison
```

```
   delta_cost delta_intervention_cost  delta_qaly      icer quadrant
        <num>                   <num>       <num>     <num>   <char>
1:  -108.3687                37.49893 0.004784424 -22650.32 dominant
   delta_ischaemic_stroke delta_haemorrhagic_stroke delta_major_bleed delta_deaths
                    <num>                     <num>             <num>        <num>
1:                  -32.7                     -7.96            -12.98        -2.15
```

- `delta_cost` and `delta_qaly`: the incremental cost and QALYs per patient.
- `delta_intervention_cost`: the part of `delta_cost` that is intervention cost.
- the other `delta_` columns: the difference in each event count and in deaths, for the whole cohort.
- `icer`: the mean incremental cost divided by the mean incremental QALYs.
- `quadrant`: where the result lies on the CE plane. It says how to read the ICER:

quadrant	iABC compared with usual care	Printed as, for example
`costlier, more QALYs`	Costs more, gives more QALYs	€12,400 per QALY gained
`dominant`	Costs less (or the same), gives more QALYs	iABC dominant (cheaper, more QALYs)
`dominated`	Costs more (or the same), gives fewer QALYs	iABC dominated (costlier, fewer QALYs)
`cheaper, fewer QALYs`	Costs less, gives fewer QALYs	€8,150 saved per QALY lost
`no QALY difference`	The same QALYs	undefined (no difference in QALYs)

The same differences for each run:

```
head(results$comparison_by_run)
```

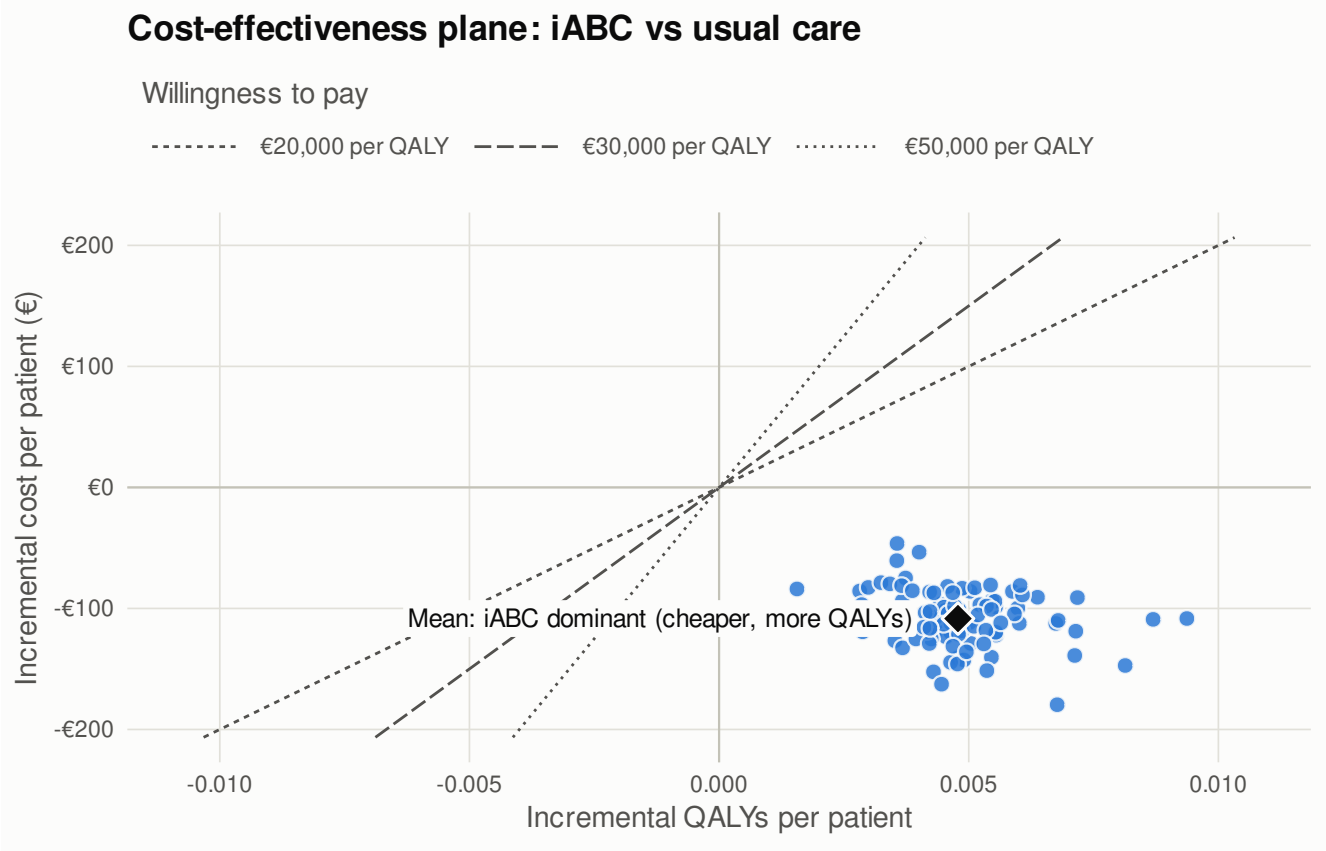
```
     run  seed delta_cost delta_intervention_cost  delta_qaly delta_ischaemic_stroke
   <int> <num>      <num>                   <num>       <num>                  <int>
1:     1  2026  -96.34138                37.44662 0.004423595                    -33
2:     2  2027 -108.60145                37.72581 0.002737998                    -24
3:     3  2028  -85.63400                37.08563 0.002815770                    -33
4:     4  2029  -82.62992                37.22436 0.002988407                    -36
5:     5  2030  -91.03477                37.57388 0.007176810                    -45
6:     6  2031 -109.03745                37.52646 0.008693408                    -37
   delta_haemorrhagic_stroke delta_major_bleed delta_deaths
                       <int>             <int>        <int>
1:                       -10                -1           -1
2:                       -12               -11            0
3:                        -2               -16           -1
4:                        -3                -8           -3
5:                         4               -11           -5
6:                        -8               -20           -9
```

```
results$comparison_by_run[, .(mean = mean(delta_qaly), lowest = min(delta_qaly), highest =
↪  max(delta_qaly))]
```

```
          mean     lowest     highest
         <num>      <num>       <num>
1: 0.004784424 0.00156039 0.009369034
```

4 The cost-effectiveness plane

```
plot(results)
```



- Each blue point is one run: incremental QALYs per patient across, incremental cost per patient up.
- The diamond is the mean across the runs, labelled with the result in the words of its quadrant.
- The dashed lines are the willingness-to-pay thresholds in `wtp`. Points below a line are cost-effective at that threshold.
- The runs use the same inputs and differ only in their random numbers, so the spread of the points shows how much a single run varies by chance.

`plot(results)` returns the plot, so it can be saved or changed with ggplot2:

```
p <- plot(results)
ggplot2::ggsave("my_ce_plane.pdf", p, width = 7, height = 5)
```

5 The output files

Every call to `tm_run()` writes these files to `outputs/`:

File	What it holds
`summary_by_run.csv`	One row per run and arm (`results$by_run`)
`summary.csv`	The mean per arm (`results$summary`)

File	What it holds
`comparison_by_run.csv`	iABC minus usual care, one row per run (`results$comparison_by_run`)
`comparison.csv`	The mean differences, the ICER and its quadrant (`results$comparison`)
`ce_plane.png`	The CE plane

```
list.files("outputs")
```

```
[1] "ce_plane.png"          "comparison_by_run.csv" "comparison.csv"
[4] "summary_by_run.csv"    "summary.csv"
```

```
fread("outputs/comparison.csv")
```

```
   delta_cost delta_intervention_cost  delta_qaly      icer quadrant
        <num>                   <num>       <num>     <num>   <char>
1:  -108.3687                37.49893 0.004784424 -22650.32 dominant
   delta_ischaemic_stroke delta_haemorrhagic_stroke delta_major_bleed delta_deaths
                    <num>                     <num>             <num>        <num>
1:                  -32.7                     -7.96            -12.98        -2.15
```